

Computing the Bundle Gradient-Norm-Squared Criterion: Structure, and a Convex–Concave Procedure

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Abstract

This note collects what we currently understand about computing

$$\text{GNS} = \max_{\lambda \in \Delta_K} \min_{i=1, \dots, m} \|\nabla F_\lambda(\theta_i)\|^2,$$

the max–min gradient-norm-squared criterion for a bundle of points under linear scalarization of K smooth objectives. The headline recommendation is that we should *not* attempt to solve this maximization globally. Our operational need is a λ at which the current bundle performs poorly, so that we can generate a new bundle point; that need is met by any feasible point of good value, and feasible points are exactly what cheap local ascent supplies. We recommend a multistart convex–concave procedure (CCP) run on the Gram matrices of the precomputed gradients. Every iterate is a valid lower bound on GNS, the iteration is monotonically ascending, and each step reduces to a small zero-sum matrix game. We also record the structural facts that motivate this choice, the exactly solvable low- K cases, off-the-shelf software options, and the approaches we considered and set aside.

1 Setting and notation

1.1 Objectives, scalarization, bundle

Let $F_1, \dots, F_K : \mathbb{R}^d \rightarrow \mathbb{R}$ be L -smooth (each ∇F_k is L -Lipschitz, with a common constant L), and not necessarily convex. Write

$$\Delta_K := \left\{ \lambda \in \mathbb{R}^K : \lambda \geq 0, \sum_{k=1}^K \lambda_k = 1 \right\}, \quad F_\lambda := \sum_{k=1}^K \lambda_k F_k.$$

Because the weights are nonnegative and sum to one, $\|\nabla F_\lambda(x) - \nabla F_\lambda(y)\| \leq \sum_k \lambda_k L \|x - y\| = L \|x - y\|$, so *every* scalarization F_λ is L -smooth with the same constant, uniformly in λ .

We are given a *bundle* $\theta_1, \dots, \theta_m \in \mathbb{R}^d$ together with all mK values $F_k(\theta_i)$ and all mK gradients $\nabla F_k(\theta_i)$.

1.2 The criterion

Define, for $i \in [m]$,

$$\phi_i(\lambda) := \|\nabla F_\lambda(\theta_i)\|^2, \quad \phi(\lambda) := \min_{i \in [m]} \phi_i(\lambda), \quad \text{GNS} := \max_{\lambda \in \Delta_K} \phi(\lambda).$$

The reading is adversarial in λ and cooperative in i : an adversary chooses the trade-off λ , and we are credited with the *best* point in the bundle for that λ ; one bundle point may serve many λ . Thus $\text{GNS} \leq \varepsilon^2$ certifies that for every $\lambda \in \Delta_K$ the bundle contains some θ_i that is ε -stationary for F_λ . Small is good. The maximum is attained, since Δ_K is compact and ϕ is continuous as a minimum of finitely many continuous functions.

Two observations about what the criterion does *not* use. The values $F_k(\theta_i)$ do not appear; nor does L . Both become relevant only downstream, when we use the computed λ^* to generate a new bundle point — a topic deliberately excluded from this note.

1.3 Reduction to Gram matrices: the ambient dimension disappears

Gradient is linear in λ . Collecting the per-objective gradients at θ_i into

$$G_i := [\nabla F_1(\theta_i) \ \cdots \ \nabla F_K(\theta_i)] \in \mathbb{R}^{d \times K}, \quad \text{we have} \quad \nabla F_\lambda(\theta_i) = G_i \lambda.$$

Proposition 1 (Dimension collapse). *Let $Q_i := G_i^\top G_i \in \mathbb{S}_+^K$, whose entries are the gradient inner products $[Q_i]_{k\ell} = \langle \nabla F_k(\theta_i), \nabla F_\ell(\theta_i) \rangle$. Then $\phi_i(\lambda) = \lambda^\top Q_i \lambda$ and*

$$\text{GNS} = \max_{\lambda \in \Delta_K} \min_{i \in [m]} \lambda^\top Q_i \lambda.$$

This is the single most important computational fact in the note. Forming $\{Q_i\}_{i=1}^m$ costs $O(mK^2d)$ once and is embarrassingly parallel; afterwards d never appears again, and we are left with an optimization over a $(K-1)$ -dimensional simplex. In our regime ($K < 10$) this is a very low-dimensional problem with many data matrices, which is why exhaustive seeding and cheap repeated local solves are affordable.

In epigraph form,

$$\text{GNS} = \max \left\{ t : t \leq \lambda^\top Q_i \lambda \ \forall i \in [m], \ \lambda \in \Delta_K \right\}, \quad (1)$$

whose constraints are *reverse-convex*. Problem (1) is therefore a DC (difference-of-convex) program, not a convex one.

Finally, write $\mathcal{A}(\lambda) := \arg \min_{i \in [m]} \phi_i(\lambda)$ for the *active set* at λ : the bundle points that attain the inner minimum.

2 Structure of the criterion

2.1 Curvature

Each ϕ_i is a convex quadratic, since $Q_i \succeq 0$. Hence $\phi = \min_i \phi_i$ is a pointwise minimum of convex functions and is in general *neither convex nor concave*, and GNS is the maximization of a nonconcave function over a polytope. Note the order matters: the transposed problem $\min_\lambda \max_i \phi_i(\lambda)$ would be a convex program. Ours is not.

2.2 Where maximizers live

Local maxima of ϕ sit at “crossings” of the lower envelope, not in the interiors of single pieces. The following makes this precise, including the degenerate exception.

Lemma 2 (Interior maximizers are crossings). *Let λ^* be a local maximizer of ϕ over Δ_K , let S be the smallest face of Δ_K containing λ^* , and let T_S denote the tangent space of S . If $|\mathcal{A}(\lambda^*)| = 1$, say $\mathcal{A}(\lambda^*) = \{i\}$, then $G_i v = 0$ for every $v \in T_S$; equivalently, ϕ_i is constant on S .*

Proof. Since $\phi_j(\lambda^*) > \phi_i(\lambda^*)$ for $j \neq i$, continuity gives a relative neighborhood U of λ^* in Δ_K on which $\phi = \phi_i$. Fix $v \in T_S$. As $\lambda^* \in \text{ri}(S)$, we have $\lambda^* \pm tv \in U \cap S$ for all small $t > 0$. Convexity of ϕ_i gives $\phi_i(\lambda^*) \leq \frac{1}{2}[\phi_i(\lambda^* + tv) + \phi_i(\lambda^* - tv)]$, while local maximality bounds each bracketed term by $\phi_i(\lambda^*)$. Hence both inequalities are equalities, so ϕ_i is affine along v ; as ϕ_i is the quadratic form $\lambda \mapsto \|G_i \lambda\|^2$, this forces $\|G_i v\|^2 = 0$. $\square$

Corollary 3. *Except in the degenerate case where some ϕ_i is constant on a face, every local maximizer lying in the relative interior of a face of dimension ≥ 1 — in particular every interior maximizer — has at least two active bundle points. The only generic single-active-point maximizers are the vertices $e_1, \dots, e_K$.*

This has two practical consequences. It tells us what a converged local solution should look like (a tie among ≥ 2 bundle points), which is a useful debugging invariant. And it explains why an ascent method that linearizes at the current point behaves well here: the iterates are naturally driven toward ties rather than stalling in the interior of a single quadratic piece.

2.3 When is GNS zero?

Proposition 4. *GNS = 0 if and only if some bundle point is simultaneously stationary for all K objectives, i.e. there exists i with $\nabla F_k(\theta_i) = 0$ for every $k \in [K]$.*

Proof. Sufficiency is immediate: such an i gives $\phi_i \equiv 0$, hence $\phi \equiv 0$. For necessity, suppose GNS = 0. Since $\phi \geq 0$, this forces $\phi \equiv 0$ on Δ_K , i.e. $\Delta_K = \bigcup_{i=1}^m Z_i$ with $Z_i := \Delta_K \cap \ker G_i$. If $G_i \neq 0$, then $\ker G_i$ cannot contain $\text{aff}(\Delta_K) = \{\lambda : \mathbf{1}^\top \lambda = 1\}$, because $e_1, \dots, e_K \in \text{aff}(\Delta_K)$ span $\mathbb{R}^K$. Hence $\ker G_i \cap \text{aff}(\Delta_K)$ is a proper affine subspace of $\text{aff}(\Delta_K)$, so Z_i has zero $(K-1)$ -dimensional Lebesgue measure relative to $\text{aff}(\Delta_K)$. Finitely many such null sets cannot cover Δ_K , which has positive relative measure. Therefore $G_i = 0$ for some i . $\square$

Operationally: the bundle-improvement loop will never drive GNS to zero unless a common stationary point is found, so any stopping rule must be stated against an absolute target $\varepsilon^2 > 0$. Which leads immediately to the next point.

2.4 Sensitivity to objective scaling

GNS is not invariant to rescaling the objectives, and the failure is not by a constant factor. This matters because the choice of normalization changes *which* λ^* we find, and hence the entire trajectory of bundle construction.

Proposition 5 (Transformation law). *Fix $c \in \mathbb{R}_{>0}^K$ and set $\tilde{F}_k := c_k F_k$, with $\widetilde{\text{GNS}}$ the resulting criterion for the same bundle. Then*

$$\widetilde{\text{GNS}} = \max_{\mu \in \Delta_K} \frac{\phi(\mu)}{(\sum_k \mu_k / c_k)^2}, \quad \text{and consequently} \quad c_{\min}^2 \text{GNS} \leq \widetilde{\text{GNS}} \leq c_{\max}^2 \text{GNS}.$$

Proof. With $C := \text{diag}(c)$ we have $\tilde{G}_i = G_i C$, hence $\tilde{\phi}_i(\lambda) = \|G_i C \lambda\|^2 = \phi_i(C \lambda)$, where ϕ_i is read as a quadratic form on all of $\mathbb{R}^K$. The map $\lambda \mapsto \mu := C \lambda / (\mathbf{1}^\top C \lambda)$ is a bijection of Δ_K onto itself, with inverse $\mu \mapsto C^{-1} \mu / (\mathbf{1}^\top C^{-1} \mu)$. Under it $C \lambda = s \mu$ with $s = 1 / (\mathbf{1}^\top C^{-1} \mu) = 1 / \sum_k (\mu_k / c_k) > 0$, so $\tilde{\phi}_i(\lambda) = s^2 \phi_i(\mu)$. The positive factor s^2 passes through the minimum over i , giving the stated identity. The sandwich follows from $\sum_k \mu_k / c_k \in [1/c_{\max}, 1/c_{\min}]$ for $\mu \in \Delta_K$, so the weight lies in $[c_{\min}^2, c_{\max}^2]$ pointwise. $\square$

So the rescaling acts as a projective reparametrization of the simplex together with a *non-constant* reweighting; the slack in the sandwich is governed by the squared condition number $(c_{\max}/c_{\min})^2$. Whatever normalization we adopt for the F_k is part of the definition of the criterion, not a free choice, and should be stated explicitly wherever GNS appears.

3 Elementary bounds

Define $A \in \mathbb{R}^{m \times K}$ by $A_{ik} := \|\nabla F_k(\theta_i)\|^2 = [Q_i]_{kk}$ — the diagonals of the Gram matrices, no cross terms. Let

$$\text{val}(A) := \max_{\lambda \in \Delta_K} \min_{i \in [m]} (A \lambda)_i = \min_{w \in \Delta_m} \max_{k \in [K]} (A^\top w)_k$$

be the value of the zero-sum matrix game with payoff A , in which the row (bundle) player minimizes. This is a linear program in $K + 1$ variables.

Proposition 6 (Free sandwich). $\max_k \min_i A_{ik} \leq \text{GNS} \leq \text{val}(A) \leq \min_i \max_k A_{ik}$.

Proof. For the left inequality, evaluate at the vertices: $\phi(e_k) = \min_i \|\nabla F_k(\theta_i)\|^2 = \min_i A_{ik}$, and $\text{GNS} \geq \phi(e_k)$ for each k . For the middle inequality, convexity of $\|\cdot\|^2$ and $\lambda \in \Delta_K$ give $\phi_i(\lambda) = \|\sum_k \lambda_k \nabla F_k(\theta_i)\|^2 \leq \sum_k \lambda_k \|\nabla F_k(\theta_i)\|^2 = (A \lambda)_i$, so $\phi \leq \min_i (A \lambda)_i$ pointwise. The right inequality is the standard pure/mixed comparison: for any λ and any fixed i_0 , $\min_i (A \lambda)_i \leq (A \lambda)_{i_0} \leq \max_k A_{i_0 k}$. $\square$

All three quantities are essentially free once the Q_i are formed. Compute them first: the outer two are pure strategies, and when the game A has a pure saddle point the sandwich closes and no further work is needed. Note also that the maximizer λ_A of the middle LP is a sensible warm start for the ascent method of Section 4, being the maximizer of a global overestimate of ϕ .

3.1 The exact convexification collapses to $\text{val}(A)$

The natural next move is to lift $\Lambda = \lambda \lambda^\top$, which renders the objective $\Lambda \mapsto \min_i \langle Q_i, \Lambda \rangle$ concave and pushes all nonconvexity into the rank-one constraint. The exact convexification replaces $\{\lambda \lambda^\top : \lambda \in \Delta_K\}$ by its convex hull

$$\mathcal{C} := \text{conv}\{\lambda \lambda^\top : \lambda \in \Delta_K\} = \{\Lambda \in \mathcal{CP}_K : \langle \mathbf{1} \mathbf{1}^\top, \Lambda \rangle = 1\},$$

where $\mathcal{CP}_K$ is the completely positive cone. This route is a dead end, and the reason is worth recording because it is not the usual standard-quadratic-programming story.

Lemma 7 (Collapse). $\max_{\Lambda \in \mathcal{C}} \min_{i \in [m]} \langle Q_i, \Lambda \rangle = \text{val}(A)$.

Proof. $\mathcal{C}$ is exactly the set of second-moment matrices $\mathbb{E}_P[\lambda\lambda^\top]$ over probability measures P on the compact set Δ_K , so by linearity the left side equals $\max_P \min_i \mathbb{E}_P[\phi_i(\lambda)]$. This is a zero-sum game with bilinear payoff over the convex strategy sets $\{P\}$ and Δ_m ; by minimax it equals

$$\min_{w \in \Delta_m} \max_P \mathbb{E}_P \left[\sum_i w_i \phi_i(\lambda) \right] = \min_{w \in \Delta_m} \max_{\lambda \in \Delta_K} \lambda^\top \left(\sum_i w_i Q_i \right) \lambda,$$

since maximizing a linear functional of P places all mass on maximizers of the integrand. For each w , the matrix $\sum_i w_i Q_i$ is positive semidefinite, so the inner maximization is that of a *convex* function over a polytope and is attained at a vertex: $\max_{\lambda \in \Delta_K} \lambda^\top (\sum_i w_i Q_i) \lambda = \max_k \sum_i w_i [Q_i]_{kk} = \max_k (A^\top w)_k$. Taking the minimum over $w \in \Delta_m$ gives $\text{val}(A)$. $\square$

Corollary 8. *Any relaxation obtained by replacing $\mathcal{C}$ with a tractable superset — the Shor relaxation, the doubly-nonnegative cone, or any outer approximation of $\mathcal{CP}_K$ — yields a bound no smaller than $\text{val}(A)$, hence no better than the linear program of Proposition 6.*

The mechanism is worth stating plainly, since it inverts the usual intuition. In a standard quadratic program the lifted objective is *linear* in Λ , so convexification is lossless. Here the lifted objective is *concave*, so its maximum over $\mathcal{C}$ need not be attained at an extreme point, and the relaxation gains exactly the value of letting the λ -player randomize. That is precisely what we must forbid: a good bundle achieves small ϕ_i through *cancellation* among the $\nabla F_k(\theta_i)$, and Jensen’s inequality — which is all $\text{val}(A)$ sees — destroys cancellation. We should expect $\text{val}(A)$ to be loose in exactly the regime we care about.

Remark 9 (What this does *not* rule out). Lemma 7 is a statement about the single lifting $\Lambda = \lambda\lambda^\top$ in the λ variables alone. It does *not* say that moment/SOS methods fail in general. The Lasserre hierarchy applied to the epigraph polynomial program (1) in the joint variables (t, λ) does converge to GNS; the Archimedean condition holds since Δ_K is compact and t can be bounded a priori using Proposition 6. Moreover the sizes are not obviously prohibitive at our scale: with $K + 1 \leq 10$ variables, the level- r moment matrix has order $\binom{K+1+r}{r}$, which is 66 at $r = 2$ and 286 at $r = 3$ for $K = 9$, with m small localizing blocks. We have not tested whether any low level is tight here, so this is flagged as an open direction rather than a recommendation — but it is the one convex-relaxation route that Lemma 7 leaves open, and it would be worth a pilot if certification ever becomes a priority.

4 The convex–concave procedure

4.1 The minorant, and why the subproblem is a matrix game

Fix a current iterate $\lambda_c \in \Delta_K$. Since each ϕ_i is convex and differentiable, it lies above its tangent plane at λ_c :

$$\phi_i(\lambda) \geq \ell_i^{(c)}(\lambda) := \phi_i(\lambda_c) + \langle \nabla \phi_i(\lambda_c), \lambda - \lambda_c \rangle = 2\lambda_c^\top Q_i \lambda - \lambda_c^\top Q_i \lambda_c, \quad (2)$$

with equality at $\lambda = \lambda_c$. Define $\psi_c := \min_i \ell_i^{(c)}$, a concave piecewise-affine minorant of ϕ that touches ϕ at λ_c . The CCP step maximizes the minorant:

$$\lambda_+ \in \arg \max_{\lambda \in \Delta_K} \psi_c(\lambda). \quad (3)$$

Because $\mathbf{1}^\top \lambda = 1$ on Δ_K , the constant in (2) can be absorbed into the linear term, so (3) is *literally a zero-sum matrix game*:

$$\lambda_+ \in \arg \max_{\lambda \in \Delta_K} \min_{i \in [m]} (M^{(c)} \lambda)_i, \quad M_{ik}^{(c)} = 2 \langle \nabla F_k(\theta_i), \nabla F_{\lambda_c}(\theta_i) \rangle - \|\nabla F_{\lambda_c}(\theta_i)\|^2. \quad (4)$$

The payoff matrix is $m \times K$: rows are bundle points, columns are objectives, and entry (i, k) is an alignment score between objective k 's gradient at θ_i and the current scalarized gradient there. As an LP, (4) reads $\max t$ subject to $t \mathbf{1} \leq M^{(c)} \lambda$, $\mathbf{1}^\top \lambda = 1$, $\lambda \geq 0$: $K + 1$ variables and m inequality rows.

Two points are worth emphasizing. First, ψ_c is concave, so the subproblem is solved *exactly* by the LP — there is no inner approximation error. Second, the LP optimum is a vertex of the epigraph polytope in (t, λ) space, *not* a vertex of Δ_K , so iterates move freely into the relative interior.

4.2 Ascent, fixed points, convergence

Lemma 10 (Monotone ascent). *Any λ_+ solving (3) satisfies $\phi(\lambda_+) \geq \phi(\lambda_c)$.*

Proof. From (2), $\psi_c \leq \phi$ pointwise on $\mathbb{R}^K$ and $\psi_c(\lambda_c) = \phi(\lambda_c)$. Since λ_c is feasible for (3), $\phi(\lambda_+) \geq \psi_c(\lambda_+) \geq \psi_c(\lambda_c) = \phi(\lambda_c)$. $\square$

We call λ_c *stationary* for $\max_{\Delta_K} \phi$ if $\phi'(\lambda_c; v) \leq 0$ for every feasible direction $v \in T_{\Delta_K}(\lambda_c)$, where $\phi'(\lambda_c; v) = \min_{i \in \mathcal{A}(\lambda_c)} \langle \nabla \phi_i(\lambda_c), v \rangle$ is the directional derivative of a minimum of smooth functions.

Lemma 11 (Fixed points are exactly the stationary points). *λ_c solves (3) if and only if λ_c is stationary for $\max_{\Delta_K} \phi$.*

Proof. Fix a feasible direction v . For $i \notin \mathcal{A}(\lambda_c)$ we have $\ell_i^{(c)}(\lambda_c) = \phi_i(\lambda_c) > \phi(\lambda_c)$, so such i are inactive for the minorant at all sufficiently small $t > 0$; for $i \in \mathcal{A}(\lambda_c)$ we have $\ell_i^{(c)}(\lambda_c) = \phi(\lambda_c)$. Hence

$$\psi_c(\lambda_c + tv) = \phi(\lambda_c) + t \min_{i \in \mathcal{A}(\lambda_c)} \langle \nabla \phi_i(\lambda_c), v \rangle \quad \text{for all small } t > 0,$$

using that each $\ell_i^{(c)}$ is affine. So λ_c is a local maximizer of ψ_c over Δ_K iff $\min_{i \in \mathcal{A}(\lambda_c)} \langle \nabla \phi_i(\lambda_c), v \rangle \leq 0$ for every feasible v , which is stationarity of ϕ . Since ψ_c is concave and Δ_K convex, local and global maximality of ψ_c coincide. $\square$

Proposition 12 (Convergence). *Let $\{\lambda_c\}_{c \geq 0} \subset \Delta_K$ be generated by (3). Then $\{\phi(\lambda_c)\}$ is nondecreasing and convergent, and every accumulation point of $\{\lambda_c\}$ is stationary for $\max_{\Delta_K} \phi$.*

Proof. Monotonicity is Lemma 10; boundedness above (e.g. by $\text{val}(A)$) gives convergence of the values to some ϕ^∞ . Let $\lambda_{c_j} \rightarrow \bar{\lambda}$ along a subsequence. Continuity of ϕ gives $\phi^\infty = \phi(\bar{\lambda})$, and since the full value sequence converges, $\phi(\lambda_{c_{j+1}}) \rightarrow \phi(\bar{\lambda})$ as well. For every $\lambda \in \Delta_K$, optimality of $\lambda_{c_{j+1}}$ for (3) gives $\phi(\lambda_{c_{j+1}}) \geq \psi_{c_j}(\lambda_{c_{j+1}}) \geq \psi_{c_j}(\lambda)$. The map $(\lambda_c, \lambda) \mapsto \psi_c(\lambda)$ is jointly continuous, so letting $j \rightarrow \infty$ yields $\phi(\bar{\lambda}) \geq \psi_{\bar{\lambda}}(\lambda)$ for all $\lambda \in \Delta_K$. As $\psi_{\bar{\lambda}}(\bar{\lambda}) = \phi(\bar{\lambda})$, this says $\bar{\lambda}$ solves (3) at $\bar{\lambda}$, and Lemma 11 applies. $\square$

4.3 Stopping rule

Lemma 11 supplies a natural and fully computable stopping criterion. Define the *predicted improvement*

$$\delta_c := \text{val}(M^{(c)}) - \phi(\lambda_c) \geq 0,$$

i.e. the optimal value of the subproblem LP minus the current value. Then $\delta_c = 0$ exactly at stationary points, and δ_c is available at no extra cost as a by-product of the step. Stop when $\delta_c \leq \tau$ for a tolerance τ chosen relative to the current value (e.g. $\tau = 10^{-8} \max\{1, \phi(\lambda_c)\}$). Note δ_c is *not* a bound on $\text{GNS} - \phi(\lambda_c)$; it is a local stationarity residual only.

4.4 The algorithm

Algorithm 1: Multistart CCP for a high-value λ

Input: Gram matrices $Q_1, \dots, Q_m \in \mathbb{S}_+^K$; seed count N ; restart count r ; tolerance τ ; iteration cap T .

Output: $\hat{\lambda}$ and the certified lower bound $\phi(\hat{\lambda}) \leq \text{GNS}$.

- 1 Form $A_{ik} \leftarrow [Q_i]_{kk}$ and compute the sandwich of Proposition 6;
- 2 **if** $\max_k \min_i A_{ik} = \text{val}(A)$ **then return** the corresponding vertex (sandwich closed);
- 3 Build a seed set Σ : the K vertices, the game maximizer λ_A , and N points drawn uniformly from Δ_K ;
- 4 Evaluate ϕ on all of Σ in one batched contraction; retain the r best, well-separated seeds;
- 5 **foreach** *retained seed* $\lambda^{(0)}$ **do**
 - 6 $c \leftarrow 0$;
 - 7 **repeat**
 - 8 Form $M^{(c)}$ via (4) ; // $O(mK^2)$
 - 9 Solve the LP $\max\{t : t\mathbf{1} \leq M^{(c)}\lambda, \lambda \in \Delta_K\}$, warm-started from the previous basis;
 - 10 $\lambda_{c+1} \leftarrow$ the LP maximizer; $\delta_c \leftarrow t^* - \phi(\lambda_c)$; $c \leftarrow c + 1$;
 - 11 **until** $\delta_c \leq \tau$ **or** $c = T$;
 - 12 Record the limit point and its value;
- 13 **return** the recorded point of largest value (and, optionally, the whole pool of distinct local maximizers);

4.5 Cost accounting

Stage	Cost	Notes
Form $\{Q_i\}$	$O(mK^2d)$ time, $O(mK^2)$ memory	once; parallel in i ; d then gone
Batch-evaluate ϕ at N seeds	$O(NmK^2)$	one tensor contraction
Form $M^{(c)}$	$O(mK^2)$	per CCP iteration
Solve subproblem LP	$K + 1$ columns, m rows	dual simplex, warm-started

For scale: $N = 10^4$ seeds, $m = 100$, $K = 9$ puts the screening pass under 10^8 flops, and each CCP iteration is a trivially small LP. Multistart with tens of restarts is cheap.

5 Initialization, multistart, and reuse

Seeds. Use the K vertices e_k (free, and they supply the lower bound of Proposition 6); the maximizer λ_A of the Jensen game; and a batch of uniform draws on Δ_K , obtained as normalized i.i.d. $\text{Exp}(1)$ vectors, or from a low-discrepancy sequence in $[0, 1]^{K-1}$ pushed onto the simplex. In dimension $K - 1 \leq 8$, a few thousand points give reasonable coverage.

Screening then polishing. Evaluate ϕ on all seeds in one batched contraction against the tensor $(Q_i)_{i=1}^m$ and polish only the top handful with CCP. This is far cheaper than running CCP from every seed and loses little, since CCP is a local method and nearby seeds converge to the same limit.

Keep the pool, not just the winner. Distinct local maximizers correspond to genuinely different trade-off regions that the current bundle handles poorly. Retaining several (deduplicated by, say, $\|\lambda - \lambda'\|_1$ together with the active sets $\mathcal{A}(\lambda)$) gives several candidate directions per outer round, which parallelizes downstream point generation.

Warm start across outer rounds. Adding one bundle point only dents the lower envelope locally, so the previous round’s pool of local maximizers is an excellent seed set for the next round — typically better than fresh random draws. Carry the pool forward and re-screen it alongside a smaller batch of new samples.

6 Exactly solvable small cases

$K = 2$. Parametrize $\lambda = (1 - s, s)$ for $s \in [0, 1]$. Each $\phi_i(s) = \|\nabla F_1(\theta_i) + s(\nabla F_2(\theta_i) - \nabla F_1(\theta_i))\|^2$ is a univariate convex parabola. On any interval where a single ϕ_i is active, ϕ is convex, so its maximum there is at an endpoint; hence the global maximizer is a crossing of two parabolas or an endpoint of $[0, 1]$. Two distinct parabolas cross at most twice, so the lower envelope is a Davenport–Schinzel sequence of order 2, has at most $2m - 1$ pieces, and can be built in $O(m \log m)$ by divide and conquer. Enumerating breakpoints then yields the exact global optimum with no solver at all. (A brute-force $O(m^2)$ enumeration of all pairwise crossings is perfectly adequate for moderate m .)

$K = 3$. The same idea becomes the lower envelope of m bivariate quadratic surfaces, of complexity $O(m^{2+\epsilon})$, with candidate maximizers at envelope vertices, along envelope edges, and on the boundary of Δ_3 . This is exact in principle, but implementing planar lower envelopes robustly is real work; we would only pursue it if a $K = 3$ instance needs a certified global answer.

Recommendation. If any part of the application has $K \leq 3$ and needs an exact answer, use the envelope approach and skip CCP there entirely. Otherwise Algorithm 1 is the workhorse.

7 Off-the-shelf software

7.1 The inner LP

The subproblem (4) is tiny and any LP solver will do: HiGHS (open source, and the default backend of SciPy’s `linprog`), GLPK, or a commercial solver (Gurobi, CPLEX, Mosek) if already licensed. Two implementation notes. Consecutive CCP iterations change only the coefficients of $M^{(c)}$, not the problem dimensions, so *warm-starting the dual simplex from the previous basis* is very effective and worth wiring up. If m ever becomes large — tens of thousands of bundle points — the matrix-game structure invites a no-regret / first-order method instead of simplex, trading exactness for an $O(1/\epsilon^2)$ approximate equilibrium; at our anticipated scale this is unnecessary.

7.2 If we want certified upper bounds

Should certification become a priority, the epigraph form (1) is a nonconvex QCQP and can be handed directly to a global solver, all of which perform spatial branch-and-bound and return a rigorous optimality gap: Gurobi with `NonConvex=2`, BARON, SCIP (open source), Couenne (open source, COIN-OR), or ANTIGONE. Model in the Q_i form of Proposition 1, never in the ambient d -dimensional gradients. Any of these is a strictly better “throw it at a solver” choice than a local NLP code.

Expectations should be calibrated, though. A simplicial branch-and-bound scheme for this problem has a bound gap governed by the Jensen gap of a quadratic on a subsimplex, which is $O(\text{diam}(S)^2 \cdot \max_i \|Q_i\|_2)$ — second-order consistent, so the scheme is sound — but reaching an absolute gap ϵ still requires on the order of $\epsilon^{-(K-1)/2}$ subsimplices in the worst case. That is pleasant at $K = 3$ and punishing near $K = 9$. This asymmetry, and not any defect of the solvers, is the reason we are prioritizing lower bounds.

7.3 Why a local NLP solver alone is the wrong default

It is tempting to hand (1) to IPOPT and be done. The objection is structural rather than numerical: GNS is a *maximum*, so any local method returns feasible points, i.e. *lower* bounds. IPOPT can therefore tell us a bundle is bad but can never certify that it is good. For our present purpose — finding a λ where the bundle underperforms — lower bounds are exactly what we want, but then CCP dominates on cost: each CCP step is a small LP with a guaranteed ascent, every iterate is already a valid bound, and a restart costs microseconds, whereas an interior-point solve is orders of magnitude more expensive per restart and produces no usable bound until it converges. With multistart being the main source of solution quality here, cost per restart is the binding constraint.

Remark 13 (Correction to an earlier claim). An earlier version of this analysis asserted that MFCQ typically fails for (1) at solutions where many reverse-convex constraints are active. That is incorrect: taking the direction $(dt, d\lambda) = (-1, 0)$ (with a small feasible perturbation of $d\lambda$ to handle active bounds) strictly increases every active constraint $\phi_i(\lambda) - t$, so MFCQ holds and multipliers remain bounded. What can genuinely degrade at solutions with large active sets is LICQ, which may slow an interior-point method but is not a correctness hazard. The case against IPOPT rests on the bound-direction and cost-per-restart arguments above, not on constraint qualifications.

8 Approaches considered and set aside

Convexification by lifting. Ruled out by Lemma 7: the *exact* convexification of the lifted problem already equals the elementary LP bound $\text{val}(A)$, so no tractable outer approximation of $\mathcal{CP}_K$ can improve on it. See Remark 9 for the one moment-based direction this leaves open.

Simplicial branch and bound. Sound, and the natural route to certified upper bounds, with the LP of Proposition 6 applied on each subsimplex as the bounding oracle (the subproblem on $S = \text{conv}\{v_1, \dots, v_K\}$ has exactly the same form after the substitution $\lambda = V\mu$, so the machinery is self-similar). Set aside for now on the cost grounds given in Section 7, and because certification is not the current priority.

Eliminating dominated bundle points. Point i is redundant when $\phi_i \geq \phi_j$ on Δ_K for some j , which holds iff $Q_i - Q_j$ is copositive. Two cheap sufficient tests ($Q_i - Q_j \succeq 0$, or $Q_i - Q_j \geq 0$ entrywise) catch most cases, and by Diananda’s theorem the copositive cone equals $\{P + N : P \succeq 0, N \geq 0\}$ for $K \leq 4$, making the test exact via a small SDP in that range. We now judge this not worth implementing: the motivation was reducing LP row count, but the subproblem LP has only $K + 1$ columns, m grows by only a few points per outer round, and the exactness of the test lapses precisely in the $K \geq 5$ regime we expect to be in.

Enumerating stationary points. At a stationary λ^* on a face S , first-order optimality reads $0 \in \text{conv}\{\nabla\phi_i(\lambda^*) : i \in \mathcal{A}(\lambda^*)\} + N_{\Delta_K}(\lambda^*)$, and Carathéodory implies that some *subset* of $\mathcal{A}(\lambda^*)$ of size at most $\dim S + 1 \leq K$ already certifies it. (This bounds the size of a minimal certificate, not $|\mathcal{A}(\lambda^*)|$ itself, which can be larger under ties.) In principle this makes the candidate set the roots of $O(m^K)$ small polynomial systems, i.e. polynomial in m for fixed K . This is theoretical reassurance that nothing pathological lurks at small K , not an algorithm; at $K = 9$ the count is vacuous.

9 Remarks for discussion

1. **This is column generation, and column generation never needed exact pricing.** The outer loop alternates between finding a λ where the bundle is weak and generating a new point for that λ . A λ at which the bundle is merely *bad* serves that purpose as well as the one at which it is *worst*. This is the whole argument for buying many cheap local ascents rather than one expensive global solve, and it should be stated explicitly wherever we describe the procedure, since a reader who thinks of GNS as a quantity to be computed exactly will find the design puzzling.
2. **Adding points never hurts.** GNS is nonincreasing in the bundle, since appending a point adds a term to the inner minimum and so lowers the envelope pointwise. The outer loop is therefore monotone by construction. Note the corresponding caveat: a lower bound computed for one bundle is *not* a lower bound for an enlarged one, so bounds must be recomputed, not carried forward.
3. **GNS = 0 is essentially unattainable** (Proposition 4): it requires a bundle point stationary for all K objectives at once. The certificate is inherently about small-but-positive values,

and the stopping rule must name an absolute target.

4. **The scale of the objectives is part of the criterion** (Proposition 5). Rescaling $F_k \mapsto c_k F_k$ reparametrizes the simplex projectively and reweights it by a non-constant factor, with slack governed by $(c_{\max}/c_{\min})^2$. It also moves the maximizer, hence changes which points enter the bundle. We should fix and state a normalization convention early.
5. **A converged solution should exhibit a tie** (Lemma 2). Interior local maximizers have at least two active bundle points, barring a degenerate flat piece. This is a cheap correctness check on any implementation, and the active set at λ^* identifies which bundle points are “responsible” for the weak region.
6. **Neither the function values $F_k(\theta_i)$ nor the smoothness constant L enters GNS.** They are needed only for the point-generation step, which is out of scope here. Worth flagging so that a reader does not go looking for where the smoothness assumption is used in this note — its only role is to guarantee that every F_λ is L -smooth uniformly in λ , which matters downstream.
7. **Open: does a low-level moment relaxation certify?** See Remark 9. This is the one convex approach that Lemma 7 does not kill, and the moment matrices are small at our K . Untested.